不定积分与定积分习题

1、 求下列一元不定积分

(1) $\int \frac{dx}{x^2}$

(2) $\int \frac{dx}{\sqrt{x}}$

(3) $\int \sqrt[m]{x^n}dx$(m,n 为常数)

(4) $\int (x^2-3x+2)dx$

(5) $\int \frac{dh}{\sqrt{gh}}$(g 是常数)

(6) $\int(2e^x + \frac{3}{x})dx$

(7) $\int x^3\sqrt{2-x^4}dx$

(8) $\int \frac{\cos\frac{3}{x}}{x^2}dx$

(9) $\int \frac{e^{4\sqrt{x}}}{\sqrt{x}}dx$

(10) $\int \frac{1}{x(1-2\ln x)}dx$

(11) $\int e^x \sin(e^x + 4)\, dx$

(12) $\int \left(\frac{3}{1+x^2} - \frac{2}{\sqrt{1-x^2}}\right) dx$

(13) $\int 3xe^x dx$

(14) $\int \sec x(\sec x - \tan x) dx$

(15) $\int cos^2 \frac{x}{2} dx$

(16) $\int \frac{dx}{1+\cos 2x}$

(17) $\int \frac{\cos 2x}{\cos x - \sin x} dx$

(18) $\int cot^2 x \, dx$

(19) $\int \cos\theta(\tan\theta + \sec\theta)d\theta$

(20) $\int \frac{3x^4+2x^2}{x^2+1} dx$

（21）$\int \frac{x}{\sqrt{2-3x^2}}dx$

（22）$\int \frac{3x^3}{1-x^4}dx$

（23）$\int \frac{\sin x}{cos^3 x}dx$

（24）$\int \frac{\sin x+\cos x}{\sqrt[3]{\sin x-\cos x}}dx$

（25）$\int cos^2 x sin^5 x\, dx$

（26）$\int sec^6 x\, dx$

（27）$\int \frac{1}{\arcsin x \cdot \sqrt{1-x^2}} dx$

（28）$\int \tan\sqrt{1+x^2} \cdot \frac{xdx}{\sqrt{1+x^2}}$

（29）$\int \frac{dx}{\sin x \cos x}$

（30）$\int xsin\, xdx$

（31）$\int \ln x \, dx$

（32）$\int xe^{-x} \, dx$

（33）$\int xtan^2 x \, dx$

（34）$\int \arcsin x \, dx$

2、 求下列一元定积分

(1) $\int_0^a (3x^2 - x + 1) \, dx$

(2) $\int_1^2 (x^2 + \frac{1}{x^4})\, dx$

(3) $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{dx}{1+x^2}$

(4) $\int_0^{\sqrt{3}a} \frac{dx}{a^2+x^2}$

(5) $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}}$

(6) $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$

(7) $\int_{-1}^{0} \frac{3x^4+3x^2+1}{x^2+1} dx$

(8) $\int_{0}^{\frac{\pi}{4}} tan^2\theta \, d\theta$

(9) $\int_{0}^{2\pi} |\sin x| \, dx$

(10) $\int_{0}^{2} f(x)dx$，其中$f(x) = \begin{cases} x+1, x \leq 1, \\ \frac{1}{2}x^2, x > 1. \end{cases}$

(11) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} cos^2u \, du$

(12) $\int_0^{\sqrt{2}} \sqrt{2-x^2}\, dx$

(13) $\int_0^1 te^{-\frac{t^2}{2}}\, dt$

(14) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cos 2x\, dx$

(15) $\int_0^{\pi} \sqrt{1+\cos 2x}\, dx$

(16) $\int_1^{+\infty} \frac{dx}{x^4}$

(17) $\int_0^{+\infty} e^{-ax}\, dx\ (a > 0)$